Adjacent Triangle Roots Carrying a Leaf and a Two-Edge Tail: A Second Rooted Supporting-Plane Inequality

Abstract

Let H be a triangle with one pendant leaf at one triangle vertex and one internally disjoint two-edge path at an adjacent triangle vertex. We prove that every graphon W of edge density $p \geq 1/2$ satisfies

$$t(H, W) \geq p^4(2p - 1) = \Phi_H(p).$$

The attachments occur at different roots, so the one-root mixed-branch theorem does not apply. A two-edge inclusion–exclusion inside the triangle reduces the density to a scalar function of d and $a = T_W d$. We prove a sharp supporting-plane inequality on the entire feasible rooted region $0 \leq a \leq d$, $a \geq d - (1 - p)$. Its $a - d^2$ term integrates to zero. This classifies the formerly open NetworkX Atlas graph 97.

1 The graph, target, and rooted densities

Let (Ω, μ) be an arbitrary probability space and let $W \colon \Omega^2 \to [0, 1]$ be symmetric and measurable. Put

$$p = \int_{\Omega^2} W(x, y)\, \mathrm{d}\mu(x)\, \mathrm{d}\mu(y), \qquad d = T_W \mathbf{1}, \qquad A = T_W d, \qquad C = T_W(d^2).$$

Let the triangle vertices be o, r, s. Attach a two-edge path $o - u - v$ at o and a leaf w at the adjacent triangle vertex r, with no other edges. Integrating the tail and the leaf first gives

$$t(H, W) = \int_{\Omega^3} W(x, y)W(x, z)W(y, z)A(x)d(y)\, \mathrm{d}\mu(x)\, \mathrm{d}\mu(y)\, \mathrm{d}\mu(z). \tag{1}$$

The graph has six vertices and six edges. Once the triangle is coloured, each new tree vertex has $x - 1$ choices after its parent has been coloured. Therefore

$$\chi_H(x) = x(x - 1)^4(x - 2), \qquad \chi(H) = 3,$$

and polynomial continuation gives

$$\Phi_H(p) = p^4(2p - 1). \tag{2}$$

The required interval is $p \in [1/2, 1]$.

2 Reduction from two triangle roots to one scalar pair

We reuse two elementary rooted facts, proving them here to keep the argument self-contained.

Lemma 2.1 (Rooted region and star compression)**.** *For almost every x, writing $d = d(x)$ and $a = A(x)$,*

$$0 \le a \le d, \qquad a \ge d - (1 - p),$$

and

$$C(x) \ge \begin{cases} A(x)^2/d(x), & d(x) > 0, \\ 0, & d(x) = 0. \end{cases}$$

Moreover,

$$\int_\Omega A \,\mathrm{d}\mu = \int_\Omega d^2 \,\mathrm{d}\mu.$$

Proof. Nonnegativity is immediate, and $A \le d$ follows from $d(y) \le 1$. The inequality $W(x,y)d(y) \ge W(x,y)+d(y)-1$ gives $A(x) \ge d(x)+p-1$. Cauchy–Schwarz under the finite measure $W(x,y)\,\mathrm{d}\mu(y)$ gives $A(x)^2 \le d(x)C(x)$ when $d(x) > 0$. If $d(x) = 0$, then $W(x,y) = 0$ for almost every y, and $A(x) = C(x) = 0$. Finally, symmetry and Tonelli give

$$\int A = \int_{\Omega^2} W(x,y)d(y)\,\mathrm{d}\mu^2 = \int d^2.$$

□

Lemma 2.2 (Adjacent-root reduction)**.** *Define*

$$h_p(d,a) = a^2 \left(\frac{a}{d} + d - 1\right)_+$$

when $d > 0$, and $h_p(0,0) = 0$. Then

$$t(H,W) \ge \int_\Omega h_p(d(x), A(x))\,\mathrm{d}\mu(x).$$

Proof. For fixed x, apply $bc \ge b+c-1$ to $W(x,z), W(y,z)$ in (1). After multiplication by $W(x,y)d(y)$ and integration in y, z, the three resulting terms are

$$d(x)A(x), \qquad C(x), \qquad -A(x).$$

The inner rooted density in (1) is also nonnegative. Hence

$$t(H,W) \ge \int A(x)\big(C(x) - (1 - d(x))A(x)\big)_+ \,\mathrm{d}\mu(x).$$

The positive-part function is nondecreasing. Lemma 2.1 gives, on $\{d > 0\}$,

$$\big(C - (1-d)A\big)_+ \ge A\left(\frac{A}{d} + d - 1\right)_+.$$

On $\{d = 0\}$ all relevant quantities vanish. The assertion follows. □

3 The adjacent-root supporting plane

For $n \geq 0$, write

$$B_i^n(x) = \binom{n}{i} x^i (1-x)^{n-i} \qquad (0 \leq i \leq n)$$

for the Bernstein basis on $[0,1]$. A polynomial expressed in this basis with nonnegative coefficients is nonnegative on the unit interval; the same holds for products of Bernstein bases on a box.

Lemma 3.1 (Adjacent-root supporting plane)**.** *Let $p \in [1/2, 1]$ and let $d, a \in [0,1]$ satisfy*

$$0 \leq a \leq d, \qquad a \geq d - (1-p).$$

Then

$$\begin{aligned} h_p(d,a) \geq\ & p^4(3-8p) + 2p^3(5p-2)d \\ & + p^2(5p-2)(a-d^2). \end{aligned} \tag{3}$$

Proof. Denote the right side by

$$L_p(d,a) = \alpha + \beta d + \gamma(a - d^2),$$

where

$$\alpha = p^4(3-8p), \qquad \beta = 2p^3(5p-2), \qquad \gamma = p^2(5p-2) > 0.$$

If $d = 0$, then $a = 0$ and $L_p(0,0) = \alpha \leq 0$, so assume $d > 0$.

Put $t = d(1-d)$. First suppose $a \leq t$. Then $h_p(d,a) = 0$. Since L_p is increasing in a, it is enough to put $a = t$. Direct expansion gives

$$L_p(d,t) = -p^2 Q_p(d),$$

where

$$Q_p(d) = 2(5p-2)d^2 + (-10p^2 - p + 2)d + 8p^3 - 3p^2.$$

Its leading coefficient is positive and

$$\operatorname{disc}_d(Q_p) = -(2p-1)^2(5p-2)(11p+2) \leq 0.$$

Thus $Q_p(d) \geq 0$, proving (3) in the zero region.

Now suppose $a \geq t$. Multiplying (3) by $d > 0$ reduces it to

$$G_p(d,a) := a^3 + a^2 d^2 - a^2 d - d L_p(d,a) \geq 0. \tag{4}$$

The feasible lower endpoint for a is

$$b_p(d) = \max\{d(1-d), d - (1-p)\}.$$

At its first branch,

$$G_p(d, d(1-d)) = dp^2 Q_p(d) \geq 0. \tag{5}$$

The second branch occurs exactly when $d \geq \sqrt{1-p}$. Put

$$C_p(d) = G_p(d, d - (1-p)).$$

For $p < 1$, C_p is a monic quartic and

$$\mathrm{disc}_d(C_p) = -4p^4(p-1)^4 D(p) < 0,$$

where, after $u = p - 1/2 \geq 0$,

$$\begin{aligned} D(p) = {} & 24000u^{16} + 123200u^{15} + 367520u^{14} + 798704u^{13} + 1276499u^{12} \\ & + 1566514u^{11} + \frac{2961123}{2}u^{10} + \frac{2049397}{2}u^9 + \frac{8200349}{16}u^8 + \frac{363635}{2}u^7 \\ & + \frac{1448591}{16}u^6 + \frac{43363}{2}u^5 + \frac{1137901}{256}u^4 + \frac{101681}{128}u^3 \\ & + \frac{57211}{512}u^2 + \frac{6845}{512}u + \frac{3175}{4096} > 0. \end{aligned}$$

Hence C_p has exactly two real roots. Since $C_p(0) = (p-1)^3 < 0$, one is negative and one is positive.

Let $s = \sqrt{1-p} \in (0, 1/\sqrt{2}]$. Exact factorization gives

$$C_p(s) = s(1-s^2)^2 R(s),$$

where

$$R(s) = -8s^6 - 10s^5 + 11s^4 + 21s^3 - 12s^2 - 9s + 5.$$

To see that $R(s) > 0$ without a numerical root estimate, put $x = 4s/3$. Our interval has $0 < x < 1$, and the degree-12 Bernstein coefficients of $R(3x/4)$ are

$$\begin{aligned} & 5, \ \frac{71}{16}, \ \frac{83}{22}, \ \frac{42887}{14080}, \ \frac{32447}{14080}, \ \frac{72073}{45056}, \ \frac{77449}{78848}, \\ & \frac{22375}{45056}, \ \frac{19}{110}, \ \frac{1951}{112640}, \ \frac{17}{5632}, \ \frac{115}{2048}, \ \frac{11}{256}. \end{aligned}$$

They are all positive. Therefore $C_p(s) > 0$, so the positive real root of C_p lies below s. It follows that

$$C_p(d) > 0 \qquad (d \geq \sqrt{1-p}). \tag{6}$$

At $p = 1$, the same conclusion follows directly from $C_1(d) = d(d-1)^2(d+5) \geq 0$.

It remains only to rule out a dip of the cubic $G_p(d, \cdot)$ below its nonnegative lower endpoint. Exact calculation gives

$$\mathrm{disc}_a(G_p) = -d^2p^2(d-p)^2 S(p,d), \tag{7}$$

where S has degree six in p and degree five in d. Put $x = 2p-1$ and $y = d$. In the product Bernstein basis $B_i^6(x)B_j^5(y)$, its coefficient matrix (s_{ij}) is

$$\begin{aligned} S(p,d) = {} & 20d^5p - 8d^5 - 60d^4p + 24d^4 + 12d^3p^3 - 4d^3p^2 + 60d^3p - 24d^3 \\ & + 1124d^2p^4 - 924d^2p^3 + 188d^2p^2 - 20d^2p + 8d^2 \\ & - 1364dp^5 + 636dp^4 + 156dp^3 - 76dp^2 \\ & + 1728p^6 - 1296p^5 + 243p^4, \end{aligned}$$

and the coefficient matrix is

$$\begin{pmatrix} 27/16 & 97/80 & 57/80 & 67/80 & 83/80 & 25/16 \\ 81/16 & 52/15 & 517/240 & 139/60 & 707/240 & 9/2 \\ 1161/80 & 11783/1200 & 7837/1200 & 1537/240 & 1919/240 & 2879/240 \\ 1593/40 & 1379/50 & 7801/400 & 3589/200 & 1053/50 & 149/5 \\ 1053/10 & 3813/50 & 8563/150 & 1523/30 & 8263/150 & 713/10 \\ 270 & 1032/5 & 2446/15 & 431/3 & 437/3 & 512/3 \\ 675 & 2727/5 & 2267/5 & 2017/5 & 1963/5 & 423 \end{pmatrix}.$$

Every entry is at least $57/80$, so $S(p,d) > 0$ on the whole box.

If $d \neq p$, (7) is negative. Thus the monic cubic $G_p(d,\cdot)$ has exactly one real root. Equations (5)–(6) show that this root is at or below the feasible lower endpoint $b_p(d)$, so $G_p(d,a) \geq 0$ for every feasible $a \geq b_p(d)$. If $d = p$, direct factorization gives

$$G_p(p,a) = (a-p^2)^2(a+3p^2-p) \geq 0.$$

This proves (4) and the lemma. □

Remark 3.2 (Exact polynomial audit)**.** *The large displayed coefficient certificates were independently reconstructed from the defining polynomials over the rational polynomial ring, not merely checked at sampled values.*

4 The graphon theorem and Atlas consequence

Theorem 4.1 (Adjacent leaf and two-edge tail)**.** *For every graphon W on an arbitrary probability space, of edge density $p \in [1/2,1]$,*

$$\boxed{t(H,W) \geq p^4(2p-1) = \Phi_H(p).}$$

Proof. By Lemma 2.2 and then Lemma 3.1,

$$\begin{aligned} t(H,W) &\geq \int h_p(d,A) \\ &\geq p^4(3-8p) + 2p^3(5p-2)\int d + p^2(5p-2)\int (A-d^2). \end{aligned}$$

Lemma 2.1 says that the last integral vanishes, while $\int d = p$. Therefore

$$t(H,W) \geq p^4(3-8p) + 2p^4(5p-2) = p^4(2p-1),$$

which is (2). □

At $p = 1/2$ the target is zero and no division by $2p-1$ occurs. At $p = 1$, $W = 1$ almost everywhere and both sides equal one. Every balanced complete k-partite graphon, $k \geq 2$, gives $d = p$, $A = p^2$, and equality in the supporting plane, hence realizes equality in the theorem.

Corollary 4.2 (Atlas 97)**.** *NetworkX Atlas graph* 97*, with graph6 code* `EYWO`*, is positive. It has order six, size six, chromatic number three, and*

$$\chi_{F_{97}}(x) = x(x-1)^4(x-2).$$

Every graphon of edge density $p \in [1/2,1]$ satisfies

$$t(F_{97},W) \geq p^4(2p-1) = \Phi_{F_{97}}(p).$$

Proof. Its NetworkX edge list is

$$02,\ 12,\ 13,\ 14,\ 24,\ 35.$$

Vertices $1,2,4$ form a triangle. The path $1-3-5$ is a two-edge tail at vertex 1, while $2-0$ is a leaf at the adjacent triangle vertex 2. There are no other edges, so Theorem 4.1 applies. □

5 Hidden-assumption audit and exact limitations

All graphon integrands are bounded, nonnegative, and measurable, so every change in integration order is justified by Tonelli–Fubini. Pointwise statements are almost-everywhere statements. The set $\{d = 0\}$ is handled without division. The proof uses ordinary homomorphism densities and does not assume a finite-step graphon, injectivity, regularity, constant degree, or a positive minimum degree.

For formal verification, the principal new statement is Lemma 3.1. Bernstein positivity may be formalized by expanding the two exact coefficient lists and using $B_i^n(x) \geq 0$ on $[0, 1]$. The graphon layer needs only the rooted region identities, Cauchy–Schwarz, the elementary two-edge inequality, monotonicity of positive part, and integration of the supporting plane. The graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph097.lean`.

The theorem requires exactly one leaf and one two-edge path at adjacent triangle roots. It does not cover attachments at all three roots, longer paths, a different number of leaves, or multiple tails. It also does not assert a general vertex-gluing theorem: the proof depends on the particular two-edge inclusion–exclusion and the exact scalar certificate above.

6 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/AdjTail.lean`, ending at `satisfiesLowerBound_97`. Theorem 4.1 is proved in full, on all of $p \in [1/2, 1]$. Two things are done differently, and one hypothesis turns out to be unnecessary.

The departure: no cubic discriminant, no Bernstein certificate. This note proves Lemma 3.1 by treating $G_p(d, \cdot) = a^3 + a^2d^2 - a^2d - dL_p$ as a cubic in a: it evaluates G_p at both branches of the feasible lower endpoint — one of which needs the degree-12 Bernstein expansion of the sextic $R(s)$ in $s = \sqrt{1-p}$ — and then rules out a dip by certifying the positivity of $\mathrm{disc}_a(G_p)$ through an 8×7 Bernstein coefficient matrix. The formalization certifies none of that. In the coordinates

$$X = a - pd, \qquad Y = d - p$$

— both of which vanish at the equality point $(d, a) = (p, p^2)$ *and* at the degenerate corner $(d, a) = (0, 0)$ — the cleared inequality is the single `ring` identity

$$a^2(a + d^2 - d) - d\,L_p(d, a) = (a - p^2)^2(a + d^2 - d) + p^2\big(2X^2 + (2d + p)XY + (7p - 2)dY^2\big). \quad (8)$$

The first summand is nonnegative exactly on the active region $a + d^2 - d \geq 0$, where (4) is what has to be proved. The bracket is nonnegative on the whole rooted region $0 \leq a \leq d \leq 1$, by a split at $d = p/3$.

Case $d \geq p/3$. The form is positive semidefinite outright:

$$8\big(2X^2+(2d+p)XY+(7p-2)dY^2\big) = \big(4X+(2d+p)Y\big)^2-q_p(d)\,Y^2, \qquad q_p(d) = 4d^2+(16-52p)d+p^2, \quad (9)$$

and $q_p \leq 0$ on $[p/3, 1]$ because q_p is a convex quadratic in d with

$$q_p(p/3) = \tfrac{p(48-143p)}{9} \leq 0, \qquad q_p(1) = p^2 - 52p + 20 \leq 0$$

on $p \geq 1/2$. This is the only place the interval endpoint $d \leq 1$ is used.

Case $d \le p/3$. Then $Y < 0$, and $0 \le a \le d$ together with $p \ge 1/2$ give $|X| \le pd$. Dropping $2X^2 \ge 0$ and bounding the cross term by $-(2d+p)pd(p-d)$ leaves the exact remainder

$$(7p-2)dY^2 - (2d+p)pd(p-d) = d(p-d)\big(6p^2 - 2p - (9p-2)d\big),$$

whose last factor is at least $p(9p-4)/3 \ge 0$ once $d \le p/3$.

So the active case is three `ring` identities and a handful of sign checks; no cubic is analysed, no Bernstein basis appears, and no square root of $1-p$ is ever formed. The inactive case follows this note exactly: L_p is increasing in a, and at $a = d(1-d)$ it equals $-p^2 Q_p(d)$, with the discriminant bound made explicit as

$$8(5p-2)Q_p(d) = \big(4(5p-2)d + (2-p-10p^2)\big)^2 + (2p-1)^2(5p-2)(11p+2).$$

One hypothesis of Lemma 2.1 is not needed. The feasibility bound $a \ge d - (1-p)$ appears nowhere in the formal proof. Both `plane_le_active` and `plane_nonpos` hold on the larger region $0 \le a \le d \le 1$ alone. The remaining parts of Lemma 2.1 are used and are pre-existing: $A \le d$ is `K4Tail.pathOp_le_degree`, $A^2 \le d\,C$ is `Broom.sq_pathOp_le`, and $\int A = \int d^2$ is `BookTail.integral_pathOp`. (The lower bound itself is available as `Broom.le_pathOp` if a later row wants it.)

No positive part, and no division by d. Lemma 2.2 is not formalized in the stated form. Instead of introducing h_p and reasoning about $(\cdot)_+$ and A^2/d, the Lean proof keeps the inner rooted density

$$E(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\, d(y)\, \mathrm{d}\mu(y)\, \mathrm{d}\mu(z)$$

as an object of its own (`AdjTail.leafTri`), proves the two-edge inclusion–exclusion in the form

$$E \ge C - (1-d)A \qquad (\texttt{AdjTail.leafTri_ge}),$$

and states the pointwise supporting-plane bound multiplied through by d (`AdjTail.mul_plane_le`). The set $\{d = 0\}$ is then an ordinary case in `plane_le_rooted` rather than a convention, and the two branches of h_p become the two branches of a case split on the sign of $a + d^2 - d$. The identity (1) is `AdjTail.homDensity_adjTail`, obtained through the generic peeling ladder of `Methods/Peeling.lean`.